

HUY NGUYEN

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EDUCATION

University of California - Santa Barbara

B.S. Electrical Engineering, Expected June 2028

Santa Barbara, CA

- **Honors/Awards:** Amazon Future Engineer (1 of 400 nationally), Elks Scholar (1 of 480 nationally), UCSB Promise Scholar (Full scholarship)
- **Coursework:** Data Structures & Algorithms, Database System, Signals & Systems, Digital Logic, Control System, Analog Circuit Analysis, E&M
- **Clubs:** Collegiate Tennis, Gaucho Rocket Project (Avionics Team), Ascend Consulting Group (SWE Consulting), Sigma Eta Pi

EXPERIENCE

Software Development Engineer Intern

June 2026 - September 2026

Amazon – Alexa

Seattle, WA

- Designed a WebRTC real-time agentic character calling system for Alexa+ to address a feature requested by 15,000+ customers
- Kept conversational latency under 1.5s by eliminating audio backpressure caused by a 4.7× rate mismatch between speech generation and playback
- Implemented live interruption handling that cleared queued audio and resumed generation when callers interrupted the character mid-sentence
- Shipped the end-to-end system to Amazon's pre-production infrastructure with lip-synced avatar rendering and sentence-level content-safety checks

Software Engineer Intern – Mobile & Product

April 2026 - June 2026

Polymarket

Remote

- Synthesized 300+ user interviews to identify onboarding barriers in Polymarket's mobile experience, where 45% of users were not trading
- Developed a real-time trading heat map and dual “Analyst vs. Casual” mobile experience tailored to users with different levels of trading expertise

Software Development Engineer Intern

June 2025 - September 2025

Amazon – Alexa

Seattle, WA

- Unified three backend services to deliver an Alexa+ privacy feature & gave customers natural-language access to their call history thru voice requests
- Built an LLM evaluation framework across 50+ test scenarios to catch inconsistent responses & reduced AI hallucinations via model fine-tuning
- Owned and deployed an end-to-end AWS proof of concept within an architecture designed to support ~50M requests/week

PROJECTS

iGauchoBack - UC Degree Roadmap | *Next.js, TypeScript, Supabase, React Flow, Upstash, Sentry*

May 2026 - Present

igauchoback.vercel.app

- Launched a UC degree-planning platform with 70+ interactive roadmaps across multiple campuses to transform traditional advising
- Built a TypeScript ingestion pipeline that converts inconsistent UC course catalogs into validated prerequisite graphs
- Hardened backend APIs with rate limiting, edge caching, and Sentry observability to keep the platform responsive and reliable under concurrent use

Orbit AFE | *JavaScript, CSS, SQL, Supabase, Leaflet, Globe.gl*

July 2026 - July 2026

orbitafe.vercel.app

- Launched Orbit in 24 hours, connecting interns across 25+ NA hubs through real-time 2D/3D presence maps with opt-in location sharing
- Architected real-time DMs, group chat, and nine-signal matching that converts and ranks potential connections from user profile data

Chromatic Guitar Tuner | *Python, SoundDevice, NumPy*

April 2026 - May 2026

- Built a real-time Python guitar tuner that analyzes live microphone audio to identify notes and determine whether they are sharp, flat, or in tune
- Designed an FFT-based pitch-detection pipeline w/ thread-safe audio buffering to separate signal from microphone capture for responsive analysis

LiDAR Home Security Detection System | *Arduino Uno R3, C++, I2C*

September 2024 - December 2024

- Built an Arduino LiDAR system scanning 180° in 1° increments to detect movement up to 170cm and placed Top-5 from a Northrop Grumman expo
- Hardened movement detection against sensor noise and failed readings using a 100cm proximity filter, adaptive baselines, and automatic retries

LEADERSHIP

Design & Make Ambassador | *Autodesk x UCSB*

February 2026 - Present

- Selected as 1 of 2 to launch Autodesk's UCSB presence, reaching 100+ students and 4+ talks across engineering, business, and design org

Vice President | *Sigma Eta Pi - Professional Fraternity*

October 2025 - September 2026

- Scaling membership from inception to 60+ active members in two quarters & established a structured 7-week founder education program

SKILLS & INTERESTS

Languages: Java, Python, C++, TypeScript, JavaScript, SQL, MATLAB, HTML/CSS

Tools/Frameworks: AWS (EC2, S3, SSM), WebRTC, REST APIs, Spring, GStreamer, ElevenLabs, Pandas, NumPy, Maven/Gradle, CI/CD, Git

Embedded & Hardware: Arduino Uno, I2C, LTspice, Oscilloscope, Soldering, Digital Multimeter, Fusion 360, Onshape, Revit

Hobbies: Speed Cubing, 4x4 Rock Crawling, Sports Photography, Sticker Collecting